

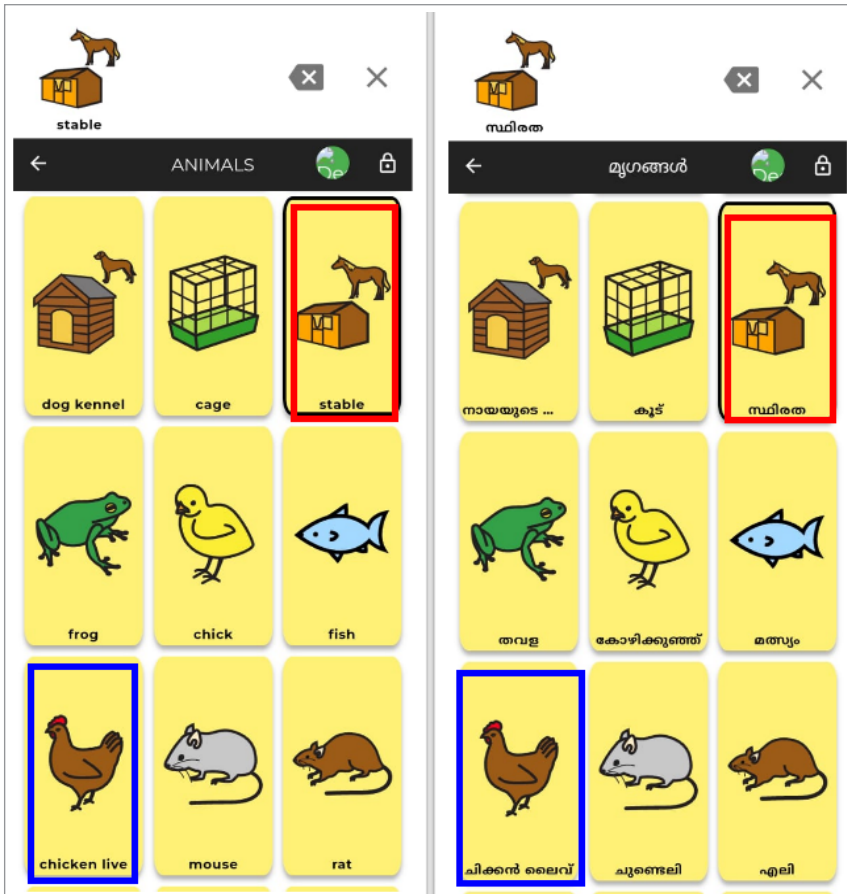


Fig. 2: A board showing images and names of animals in English and Malayalam

communication boards, picture exchange communication systems, eye gaze boards, and so on, which are priced under a few thousand rupees. High-tech AAC devices include speech-generating devices, tablets with AAC apps, and dedicated communication devices, with prices ranging from a few thousand rupees to millions of rupees.

When I first learned about AAC devices used for training in many developed countries, I was saddened to recall that none of the ten or so special schools I visited in the past two decades had anything close to these devices, despite the availability of inexpensive low-tech AAC devices in the market. All they had were charts with pictures of fruits, animals, etc, hanging in the classrooms. Furthermore, the computers

(at most three, often slow desktops) they had were used mostly for photo ops. I saw photos of students sitting in front of computers, looking confused at their names displayed in large font in Microsoft Word, typed by someone else.

However, this lack of technology in special education could be easily overcome with free and open source software, as many such programs can emulate the specialised hardware of an AAC device on even low-end laptops. The adoption of such software would transform special education in India.

Cboard: Designed for special education

Cboard is one such highly useful free and open source software, designed as an augmentative and alternative

communication (AAC) software. It allows users with speech and language impairments to communicate using symbols and text-to-speech (TTS). This cross-platform software is licensed under the MIT License. The GitHub repository for Cboard is available at <https://github.com/cboard-org>, and the English version of the Cboard information website is available at www.cboard.io/en.

Cboard offers a wide variety of graphical images for users to choose from, enabling them to construct fairly complex sentences by selecting these images. The application then reads out the sentences in the user's chosen language. Cboard supports both web and mobile platforms.

In my opinion, a touchscreen laptop is the ideal device for training with Cboard, as individuals with neurodevelopmental disabilities often struggle with poor eye-hand coordination and motor skills. However, a standard laptop or smartphone can also be used for training with Cboard. Fig. 1 shows a screenshot of the Settings page in the Cboard app, where you can change the language for displaying and reading text, among other options.

What surprised me when I first encountered the Cboard software was the lack of Malayalam language support. This was unexpected, considering that Kerala is a state known for excelling in education and human resource development in India. Despite this, Malayalam was not supported until a group of volunteers, including myself, made a request. The Cboard development team responded promptly, and Malayalam support was added. As of writing this article, the following Indian languages are supported: Bengali, Hindi, Malayalam, and Nepali.

It was also surprising that versions of Cboard in widely spoken languages such as Tamil, Telugu, Kannada, Odia, Gujarati, and